

Operators

Operators are special symbols that perform operations on variables and values - math, comparison, or logic.

How to use:

Java has several categories:

- * Arithmetic Operators - math operations.

Ex: `int a=10, b=3;`

`a+b;`, `a-b;`, `a*b;`, `a/b;`, `a%b;`

- * Relational (Comparison) Operators - return true/false

`a > b`, `a < b`, `a == b`, `a != b`, `a >= b`, `a <= b`

- * Logical Operators - Combine boolean conditions.

`boolean x=true, y=false;`

`x & y;` // true only if both true

`x || y;` // true if at least one true

`!x;` // flips the value.

- * Assignment Operators - assign or update values.

`int c=5;`

`c += 2;`

`c -= 2;`

`c *= 2;`

`c /= 2;`

- * Unary Operators - act on a single operand

`int n=5;`

`n++;`

`n--;`

`-n;`

When to use:

- * Arithmetic operators (calculations)

- * Comparison Operators (checking conditions)

- * Logical operators (Combining multiple conditions)

Where to use:

Inside conditions (if statements), loops, calculations, anywhere you need to compute or compare values.

Ex: `if (age >= 18) { } // relational + logical`

`total = price * qty; // arithmetic`

`count++; // unary`

Why we have different categories of operators, not just one type?

Without these distinct categories, you couldn't build any real logic - every if statement, loop condition, or validation check depends on operators.

Real-life Example:

- Like tools in a toolbox - hammer (arithmetic) for building, measuring tape (comparison) for checking, Combination lock logic (logical) for checking multiple conditions together. Each tool has its specific task.